Electric and magnetic fields in media

1. Derive Gauss's law for dielectrics,

$$\int_{\partial V} \vec{ds} \cdot \vec{D} = Q_{\text{ext}},$$

where Q_{ext} is the total external charge contained in the volume V.

2. Calculate the electric field and the electric potential of a charge q placed in the middle of a dielectric sphere of radius a and permittivity ϵ. Compute the discontinuities and continuities of $\vec{E}$ and $\vec{D}$ across the boundary of the sphere. Determine the polarization field $\vec{P}$ and the corresponding surface charge density σ_P.

3. Consider an infinite dielectric composed of two media with permittivities ϵ_1 and ϵ_2, separated by a plane. Find the electric field and potential of a point charge q placed in the medium ϵ_1 at a distance d from the plane. Calculate the density of polarization charges on the plane.

4. A *capacitor* is an electric device made of two conducting surfaces (the *electrodes*) separated by a dielectric medium (the *insulator*), and designed to store opposite electric charges. It is characterised by its *electric capacity* C (expressed in *Farads* in the international system), which indicates the charge Q accumulated for a given tension V between the surfaces (relation $Q = CV$). For the following configurations :

   - A conducting sphere of radius R in vacuum
   - Two concentric conducting spheres of radii a and b ($a < b$)
   - Two coaxial conducting cylinders of radii a and b ($a < b$)
   - Two parallel plane conducting surfaces of surface S, separated by a distance d

   Neglecting the edge effects and noting Q the charge accumulated on the positive electrode and ϵ the permittivity of the dielectric medium between the two surfaces, give the expression of the electric field inside and outside the capacitor and deduce their respective electric capacities.

5. Consider a dielectric sphere with uniform polarization $\vec{P}$. It can be represented as two positively and negatively charged spheres slightly displaced with respect to one another. Using this representation, calculate the surface charge density produced by the polarization. Compare to that given by the relation

$$\sigma_P = \vec{P} \cdot \vec{n}$$

where $\vec{n}$ is the unit vector normal to the sphere.

6. Calculate the potential of a dielectric sphere of radius a, polarized uniformly in the direction z with the magnitude P.

7. An *inductance* is an electric device designed to store magnetic energy and block high frequency alternative currents. It is characterised by its *electric inductance* L (expressed in *Henry* in the international system), which indicates the magnetic flux Φ accumulated for a given current I (relation $\Phi = LI$). Consider a solenoid of length $\mathcal{L}$, cross section A, with N turns and filled with a medium of permeability μ. Calculate the magnetic field generated when a current I flows through it, and deduce the value of its inductance.

8. Find the magnetic field $\vec{H}$ and magnetic induction $\vec{B}$ of a uniformly magnetized sphere of radius a with magnetization $\vec{M}$ parallel to the z-axis.

9. Consider a finite cylinder of length L and radius a, placed parallel to the z-axis and with uniform magnetization $\vec{M} = M\vec{e}_z$. Calculate the magnetic field $\vec{H}$ everywhere on the z-axis. Consider the limit $L \to \infty$ (with a fixed) to find the magnetic field $\vec{H}$, the magnetic induction $\vec{B}$ and the vector potential $\vec{A}$ everywhere in space. What is the value of the magnetization current $\vec{J}_M$?
10. A sphere of radius a and magnetic permittivity μ is inserted into a uniform magnetic field $\vec{B}_0$. Calculate the magnetization of the sphere, the magnetization currents, the resulting magnetic field and magnetic induction everywhere in space

Solutions

2. Surface charge density $\sigma_p = \frac{q}{4\pi a^2}\frac{(\epsilon-\epsilon_0)}{\epsilon}$
3. Surface charge density $\sigma_P = \frac{q}{2\pi}\frac{1}{\epsilon_1+\epsilon_2}\frac{d}{(x^2+y^2+d^2)^{3/2}}\frac{\epsilon_0}{\epsilon_1}(\epsilon_2-\epsilon_1)$
4. — Sphère dans le vide : $C = 4\pi\epsilon_0 R$
   — Condensateur sphérique : $C = \frac{4\pi\epsilon ab}{b-a}$
   — Condensateur cylindrique : $C = \frac{2\pi\epsilon\mathcal{L}}{\ln(b/a)}$
   — Condensateur plan : $C = \frac{S\epsilon}{d}$
6. $\phi(r,\theta) = \begin{cases} \frac{P}{3\epsilon_0} r\cos\theta & (r \le a) \\ \frac{Pa^3}{3\epsilon_0} r^{-2}\cos\theta & (r > a) \end{cases}$
7. Inductance $L = \mu N^2 A/\mathcal{L}$
8. $\vec{H} = \begin{cases} -\frac{M}{3}\vec{u}_z & (r \le a) \\ \frac{M}{3}\frac{a^3}{r^3}(2\cos\theta\vec{u}_r + \sin\theta\vec{u}_\theta) & (r > a) \end{cases}$
9. $H_z(0,\theta,z) = -\frac{M}{2}\left[\frac{z-L/2}{\sqrt{a^2+(z-L/2)^2}} - \text{sign}(z-L/2) - \frac{z+L/2}{\sqrt{a^2+(z+L/2)^2}} + \text{sign}(z+L/2)\right]$
10. $M = \dfrac{3(\mu-\mu_0)}{\mu_0(\mu+2\mu_0)}B_0$; $\vec{J}_M = \vec{0}$; $\vec{H} = \begin{cases} -\frac{3}{\mu+2\mu_0}\vec{B}_0 & (r \le a) \\ \frac{1}{\mu_0}\vec{B}_0 + 2\frac{\mu-\mu_0}{\mu_0(\mu+2\mu_0)}\frac{a^3}{r^3}\vec{B}_0 & (r > a) \end{cases}$

Charged particle in electric and magnetic fields

1. The non-relativistic equations of motion of a 3-dimensional harmonic oscillator have the form
$$m\frac{d^2}{dt^2}\mathbf{x} = -k\mathbf{x},$$
where $\mathbf{x}$ is the position of the oscillator, m is its mass and k is a constant.
   (a) Solve these equations and determine the frequency of oscillations ω_0
   (b) Suppose the oscillator has an electric charge e and is placed in a uniform magnetic field $\mathbf{B} = (0, 0, B)$. Write down the equations of motion. Solve them and determine the frequencies of oscillations in the presence of the magnetic field

2. Study the motion of a charged particle of electric charge e and mass m in constant, uniform and parallel electric and magnetic fields $\mathbf{E} \parallel \mathbf{B}$.
   (a) Write down and solve the equations of motion in the non-relativistic limit. To what extent does this limit remain valid ?
   (b) Do the same exercise in the relativistic case and show that, for appropriate integration constants, the solution in parametric form is
$$\begin{cases} x = AR\sin\phi \\ y = AR\cos\phi \\ z = \frac{R}{\rho}\sqrt{1+A^2}\cosh(\rho\phi) \\ ct = \frac{R}{\rho}\sqrt{1+A^2}\sinh(\rho\phi) \end{cases}$$
   where $R = mc/eB$, $\rho = E/cB$, A is an arbitrary constant and ϕ is a parameter.
   (c) Compare your solutions in the non-relativistic limit and show that they are identical

3. Consider a particle with an electric charge q moving at velocity $\vec{v}$ parallel to an electric wire in which an electric current I circulates in the same direction. The magnetic field around the wire acts on the particle and curves its trajectory. Turning into the reference frame of the particle, according to the classical expression of the Lorentz force $\vec{F} = q\vec{v} \times \vec{B}$, the force should then vanish since the particle is at rest. In this exercise we will solve this apparent paradox.
   Start by computing the electromagnetic 4-force f^μ applied on the particle with the relation
$$f^\mu = qF^\mu{}_\nu u^\nu = q\eta_{\alpha\beta}F^{\mu\alpha}u^\beta \tag{1}$$
   and evaluate it in the nonrelativistic limit. Then compute this force f'^μ in the reference frame of the particle and show that it remains unchanged (provided the force and the boost velocity are orthogonal).
   Compute the Faraday tensor $F'^{\mu\nu}$ in the new reference frame [1], and use (1) to show that it gives the right 4-force f'^μ.

1. Reminder : a rank-2 contravariant tensor $T^{\mu\nu}$ transforms as
$$T'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta T^{\alpha\beta}$$

4. Study the motion of a charged particle of electric charge e and mass m in constant, uniform and mutually orthogonal electric and magnetic fields : $\mathbf{E} \cdot \mathbf{B} = 0$, with $|\mathbf{E}| = c|\mathbf{B}|$.

   (a) Write down and solve the equations of motion in the non-relativistic case. To what extent does this limit remain valid ?

   (b) Do the same exercise in the relativistic case.

   (c) Compare your solutions in the non-relativistic limit and show that they are identical.

5. Consider the motion of a particle in a magnetic field with the magnetic field lines which are circles of radius R (the z-component of the field is zero). Derive the expression for the drift velocity $\mathbf{v}_C$,

$$\mathbf{v}_C = \frac{v_\parallel^2}{\omega_B R} \left(\frac{\mathbf{R} \times \mathbf{B}}{RB} \right),$$

   by solving the equations of motion for this particle.

   Hint : Write the Lorentz equation in cylindrical coordinates where the only non-zero component of $\mathbf{B}$ is $B_\phi = BR/\rho$ for constant curvature radius R and ρ being the radial coordinate. Look for solutions which are small oscillations around the field lines.

6. Starting from the general expression for the drift velocity

$$\mathbf{v}_D = \frac{1}{\omega_B R} \left(v_\parallel^2 + \frac{1}{2} v_\perp^2 \right) \frac{\mathbf{R} \times \mathbf{B}}{RB}$$

   show that numerically the magnitude of the drift velocity of particles of a plasma at temperature T in the magnetic field of strength B and characteristic curvature of field lines R is

$$v_D = 172 \left(\frac{T}{1\mathrm{K}} \right) \left(\frac{1\mathrm{m}}{R} \right) \left(\frac{1\mathrm{G}}{B} \right) \frac{\mathrm{cm}}{\mathrm{s}}$$

7. The magnetic field of the Earth can be represented approximately by a magnetic dipole of magnetic momentum $M = 8 \times 10^{22} \mathrm{Am}^2$ pointing towards the South. Let us study the behavior of energetic electrons in the vicinity of the Earth.

   (a) Show that the equation for a magnetic field line above the equator is $r = r_0 \sin^2 \theta$ with θ the angle measured with respect to the axis of the dipole. Fine the amplitude of the magnetic field B along a field line.

   (b) A positively charged particle revolves around a field line in the equatorial plane, with a gyration radius a and at a mean distance R from the center of the Earth. Show that the azimuthal position φ of the particle (i.e. its longitude) changes approximately linearly with time, according to

$$\varphi(t) = \varphi_0 - \frac{3}{2} \left(\frac{a}{R} \right)^2 \omega_B (t - t_0),$$

   where ω_B is the gyration frequency for a radius R.

Solutions

1. a) $\omega_0 = \sqrt{k/m}$

b) Frequencies $\omega_\pm = \dfrac{\omega_B \mp \sqrt{\omega_B^2 + 4\omega_0^2}}{2}$ with $\omega_B = eB/m$

2. a) Nonrelativistic motion :

$$\begin{cases} x(t) &= \frac{v_{x,0}}{\omega_B}\cos\omega_B t \\ y(t) &= \frac{v_{y,0}}{\omega_B}\sin\omega_B t \\ z(t) &= \frac{eE}{2}t^2 \end{cases} \tag{2}$$

b) Relativistic motion :

$$\begin{cases} x(t) = \dfrac{p_0}{eB}\sin\left(\dfrac{cB}{E}\text{arcsinh}\left(\dfrac{eEc}{\mathcal{E}_0}t\right)\right) \\ y(t) = \dfrac{p_0}{eB}\cos\left(\dfrac{cB}{E}\text{arcsinh}\left(\dfrac{eEc}{\mathcal{E}_0}t\right)\right) \\ z(t) = \sqrt{\mathcal{E}_0^2/e^2E^2 + c^2t^2} \\ \mathcal{E}(t) = \sqrt{\mathcal{E}_0^2 + (ceEt)^2} \end{cases}$$

4. a) Nonrelativistic motion :

$$\begin{cases} x(t) &\simeq x_0 + \dfrac{eE}{2m}t^2 \\ y(t) &= y_0 \\ z(t) &\simeq z_0 + \dfrac{e^2E^2}{6m^2c}t^3 \end{cases}$$

b) Relativistic motion :

$$\begin{cases} x(t) &= p_x^2/2meB \\ y(t) &= y_0 \\ z(t) &= p_x^3/6eBm^2c^2 \\ p_x^3 + 6m^2c^2p_x &= 6eBm^2c^3t \end{cases}$$

5. With a Taylor expansion around the circle of radius R :

$$\begin{cases} r(t) &= \dfrac{v_{\rho,0}}{\Omega_0}\sin\Omega_0 t + \dfrac{v_{\theta,0}^2}{R\Omega_0^2}(1-\cos\Omega_0 t) \\ \dot\theta(t) &= \dfrac{Rv_{\theta,0}}{\rho^2(t)} \simeq \dfrac{v_{\theta,0}}{R}(1-2r/R) \\ \dot z(t) &= \omega_B R\ln(1+r/R) \simeq \omega_B r \end{cases}$$

Electromagnetic waves

1. Show that a general spherically symmetric solution of the d'Alembert equation

$$\left(\frac{1}{c^2}\partial_t^2 - \boldsymbol{\nabla}^2\right) f(t,r) = 0 \tag{1}$$

has (for $r \neq 0$) the form

$$f(t,r) = \frac{1}{r}\left(f_1(ct-r) + f_2(ct+r)\right), \tag{2}$$

where $f_{1,2}$ are two arbitrary functions.

2. Show that solutions to the d'Alembert equation (1) analogous to (2) (perhaps with a different function of r multiplying the arbitrary functions f_1 and f_2) do not exist in problems having cylindrical, rather than spherical symmetry.

3. Let the magnetic field of a monochromatic plane wave be

$$\mathbf{B}(t,\mathbf{x}) = B_0 \boldsymbol{\epsilon}_x \cos(-\omega t + kz).$$

Calculate the electric field, the Poynting vector and the energy density in the wave.

4. Let the electromagnetic potentials have the form

$$\begin{aligned}&\phi(t,\mathbf{x}) = 0,\\ &\mathbf{A}(t,\mathbf{x}) = (0,0,F(\omega t - \mathbf{kx})), \qquad \omega = c|\mathbf{k}|.\end{aligned}$$

Assuming that F is such that $\nabla \cdot \mathbf{A} = 0$, calculate the corresponding electric and magnetic fields, the Poynting vector and the energy density.

5. Suppose that in a *dispersive medium* the frequency ω depends non-trivially on the wavenumber k as $\omega = \omega(k)$. In realistic light sources, a beam has component waves of several frequencies and we consider the case when these are peaked around a certain wavenumber k_0. Show that the following wave packet is a solution to the wave equation in $1+1$ dimensions

$$f(t,x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} A(k) e^{-i\omega(k)t + ikx} dk,$$

when a given plane wave mode k propagates with its *phase velocity* $v_p = \frac{\omega(k)}{k}$. The aim here is to compute the *group velocity* with which the peak of the wave packet propagates. If the dispersion is not too strong one can approximate

$$\omega(k) = \omega(k_0) + \left.\frac{d\omega(k)}{dk}\right|_{k=k_0}(k-k_0) + \ldots.$$

Use this to derive the group velocity of the wave packet $v_g = \left.\frac{d\omega(k)}{dk}\right|_{k=k_0}$.

6. A plane monochromatic electromagnetic wave of the form

$$\mathbf{E} = \mathrm{Re}\left[\mathbf{E}_0 \mathrm{e}^{-i\omega t + i\mathbf{kx}}\right],$$

where $\mathbf{E}_0$ is the complex amplitude, is totally reflected from a plane mirror placed at $x = 0$. Knowing the total energy density of the wave and assuming the relativistic relation between the energy E and momentum P, $E = cP$, calculate the pressure acting on the mirror. Knowing that the energy flux from the Sun at the Earth's position is about 1.7 kW/m^2, estimate the pressure of sunlight.

7. Consider two semi-infinite dielectrics with electric permeabilities ϵ and ϵ' and magnetic permittivities $\mu = \mu'$. For a plane incoming wave falling normally on the interface (angle of incidence $\alpha = 0$), calculate the energy flux in the incoming, reflected and refracted waves. Check energy conservation.
8. The same as the previous problem, but for the wave incident at an angle θ with respect to the normal to the surface. Consider the two polarizations separately.
9. Consider a *Fabry-Perot interferometer*, made of two parallel and identical semi-reflecting surfaces, separated by a distance L, with transmission and reflection coefficients t and r (suppose for simplicity that $t, r \in \mathbb{R}$). Calculate the *transfer function* $T_{FP} = |E_t|^2/|E_i|^2$ for an incident electromagnetic wave propagating perpendicularly to the surface. Express it in terms of the *finesse* $F = 4r^2/(1-r^2)^2$. How can you interpret this parameter ?

Solutions

3. $\vec{E} = -cB_0\vec{u}_y \cos(-\omega t + kz)$; $\vec{S} = \frac{c}{\mu_0} B_0^2 \vec{u}_z \cos^2(-\omega t + kz)$; $w = \frac{B_0^2}{\mu_0}\cos^2(-\omega t + kz)$
4. $\vec{E} = -\omega\vec{u}_z F'(\omega t - \vec{k}\cdot\vec{x})$; $\vec{B} = -k_y\vec{u}_x F'(\omega t - \vec{k}\cdot\vec{x}) + k_x\vec{u}_y F'(\omega t - \vec{k}\cdot\vec{x})$; $\vec{S} = F'^2(\omega t - \vec{k}\cdot\vec{x})\frac{\omega}{\mu_0}\vec{k}$; $w = F'^2(\omega t - \vec{k}\cdot\vec{x})\frac{k^2}{\mu_0}$
6. Pressure $\mathcal{P} = \dfrac{2\phi}{c} = \epsilon_0 E_0^2 = 1.1 \times 10^{-5}\text{Pa}$
7. $|S_T| = \frac{n'}{n}\frac{4n^2}{(n+n')^2}|S_I|$; $|S_R| = \frac{(n'-n)^2}{(n+n')^2}|S_I|$
9. $T_{FP} = \frac{1}{1+\mathcal{F}\sin^2 kL}$

Emission of electromagnetic waves

1. *Birkhoff's theorem (in electromagnetism)* : Any charge-free spherically symetric solution of the Maxwell equations is necessarily static. Prove this theorem.
   (a) Express the most general spherically symmetric electric and magnetic fields
   (b) Compute their curls and inject these into the Maxwell equations

   Hint : The curl operator in spherical coordinates reads

$$\vec{\nabla}\times\vec{A} = \frac{1}{r\sin\theta}\left(\frac{\partial(A_\varphi \sin\theta)}{\partial\theta} - \frac{\partial A_\theta}{\partial\varphi}\right)\vec{u}_r + \left(\frac{1}{r\sin\theta}\frac{\partial A_r}{\partial\varphi} - \frac{1}{r}\frac{\partial(rA_\varphi)}{\partial r}\right)\vec{u}_\theta + \frac{1}{r}\left(\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial\theta}\right)\vec{u}_\varphi$$

2. Calculate the power, averaged per period, emitted by a point charge oscillating along the z-axis, $z(t) = z_0\cos\omega t$, $x = y = 0$. Discuss the dependence of the emitted radiation on the direction at large distances from the oscillating charge, $r \gg z_0$.
3. Calculate the power emitted by an electron turning on a circular orbit

$$z = 0; \quad x = a_0\cos\omega t; \quad y = a_0\sin\omega t,$$

   a_0 being the radius of the orbit. Supposing that this is an electron on the first Bohr orbit of a hydrogen atom, estimate the time needed for the electron to fall on the nucleus.
4. The expression for the total emitted power,

$$I = \frac{\mu_0 e^2}{6\pi c}a^2, \tag{1}$$

   where a is the particle acceleration, was derived in the non-relativistic approximation and is not applicable to the case when the emitting particle is relativistic. By making the Lorentz transformation to the frame where particle is at rest (but has a non-zero acceleration) and back, derive the generalization of eq. (1) to the relativistic case,

$$I = \frac{\mu_0 e^2}{6\pi c}\frac{a^2 - \frac{1}{c^2}(\vec{v}\times\vec{a})^2}{\left(1-\frac{v^2}{c^2}\right)^3},$$

   where $\vec{a}$ is the particle acceleration and $\vec{v}$ is the particle velocity in the original frame.
5. The linear accelerator at SLAC has the length of 1.45 km. By making use of the results of the previous problem, calculate the energy emitted by an electron accelerated to 50 GeV by a constant force.

Solutions

2. $\dfrac{dE}{dt} = \dfrac{\mu_0 e^2}{6\pi c}\langle a^2\rangle$ with $a^2 = z_0^2\omega^4\cos^2\omega t$
5. Radiated energy $E_r = \frac{\mu_0 e^2}{6\pi}\frac{E_c}{mL}\sqrt{\gamma_F^2 - 1} = 6.4\text{meV}$ with $\gamma_F = 1 + \frac{E_c}{mc^2}$